

69. Clearly, 143 is divisible by 11. So, 143 is not prime.
289 is divisible by 17. So, 289 is not prime.
117 is divisible by 3. So, 117 is not prime.
359 < (20)² and prime numbers less than 20 are 2, 3, 5, 7, 11, 13, 17, 19.
And, 359 is not exactly divisible by any of them. Hence, 359 is a prime number.
70. Putting $n = 1, 2, 3, 4$ respectively in $(2n + 1)$ we get:
 $(2 \times 1 + 1) = 3$, $(2 \times 2 + 1) = 5$, $(2 \times 3 + 1) = 7$ and $(2 \times 4 + 1) = 9$,
where 3, 5, 7 are prime numbers.
 $\therefore$ The smallest value of n for which $(2n + 1)$ is not prime, is $n = 4$.
71. Clearly, 100 is divisible by 2. So, 100 is not prime.
 $(101) < (11)^2$ and prime numbers less than 11 are 2, 3, 5, 7. Clearly, 101 is not divisible by any of 2, 3, 5 and 7.
Hence, 101 is the smallest 3-digit prime number.
72. Each one of 112, 114, 116, 118 is divisible by 2. So, none is prime.
Each one of 111, 114, 117 is divisible by 3. So, none is prime.
Clearly, 115 is divisible by 5. So, it is not prime.
Each one of 112 and 119 is divisible by 7. So, none is prime.
Hence, there is only 1 prime number between 110 and 120, which is 113.
73. Let the given prime numbers be p, q, r and s . Then
 $p \times q \times r = 385$ and $q \times r \times s = 1001$
$$\Rightarrow \frac{p \times q \times r}{q \times r \times s} = \frac{385}{1001} = \frac{5}{13} \Rightarrow \frac{p}{s} = \frac{5}{13} \Rightarrow p = 5 \text{ and } s = 13.$$

Hence the largest of these prime numbers is 13.
74. Let the required prime numbers be $x, y, y + 36$. Then,
 $x + y + y + 36 = 100 \Rightarrow x + 2y = 64$
Let $x = 2$. Then, $2y = 62 \Rightarrow y = 31$.
So, these prime numbers are 2, 31 and 67.
In given choices 67 is the answer.
75. $\sqrt{437} > 20$
All prime numbers less than 20 are 2, 3, 5, 7, 11, 13, 17, 19. 161 is divisible by 7, 221 is divisible by 13 and 437 is divisible by 19. 373 is not divisible by any of the above prime numbers.
 $\therefore$ 373 is prime.
76. The required arithmetic sequence of five prime numbers is 5, 11, 17, 23, 29 and therefore, the required 5th term is 29.
77. Each of the numbers 302, 303, 304, 305, 306, 308, 309, 310, 312, 314, 315, 316 and 318 is clearly a composite number.
Out of 307, 311, 313, 317 and 319 clearly every one is prime.
Hence, there are 5 prime numbers between 301 and 320.
78. Clearly, $3 \neq 4n + 1$ and $3 \neq 4n + 3$ for any natural number n .
 $\therefore$ Statement (1) is false.
Putting $p = 3, 5, 7, 11, 13, 17$ etc. we get:
 $(p - 1)(p + 1) = (2 \times 4), (4 \times 6), (6 \times 8), (10 \times 12), (12 \times 14), (16 \times 18)$ etc., each one of which is divisible by 4.
 $\therefore$ Statement (2) is true.
Hence, (1) is false and (2) is true.
79. $n = 1 \Rightarrow (n^2 + n + 41) = (1 + 1 + 41) = 43$, which is prime.
 $n = 10 \Rightarrow (n^2 + n + 41) = (100 + 10 + 41) = 151$, which is prime.
 $n = 20 \Rightarrow (n^2 + n + 41) = (400 + 20 + 41) = 461$, which is prime.
 $n = 40 \Rightarrow (n^2 + n + 41) = (1600 + 40 + 41) = 1681$, which is divisible by 41.
Thus, 1681 is not a prime number.
Hence $n = 40$ for which $(n^2 + n + 41)$ is not prime.

80. $X_2 = (2 \times 3) + 1 = 7$, which is prime and $X_2 + 1 = 8$, which is even.
 $X_3 = (2 \times 3 \times 5) + 1 = 31$, which is prime and $X_3 + 1 = 32$, which is even.
 $X_4 = (2 \times 3 \times 5 \times 7) + 1 = 211$, which is prime and $X_4 + 1 = 212$, which is even and so on.
Thus X_k is prime and $(X_k + 1)$ is even.
Hence, 1 and 3 are true statements.
81. $6 \times 3(3 - 1) = 6 \times 3(2) = 6 \times 6 = 36$.
82. Given Expression = $(1234 + 2345 + 4567) - 3456$
 $= (8146 - 3456) = 4690$.

$$\begin{array}{r} 1234 \\ 2345 \\ + 4567 \\ \hline 8146 \\ - 3456 \\ \hline 4690 \end{array}$$

83. Let $5566 - 7788 + 9988 = x + 4444$. Then
 $(5566 + 9988) - 7788 = x + 4444$
 $\Rightarrow 15554 - 7788 = x + 4444 \Rightarrow x + 4444 = 7766$
 $\Rightarrow x = (7766 - 4444) = 3322$.

$$\begin{array}{r} 7766 \\ - 4444 \\ \hline 3322 \end{array}$$

84. Given Expression = $38649 - (1624 + 4483)$
 $= (38649 - 6107) = 32542$.
85. Given Expression = $884697 - (773697 + 102479)$
 $= 884697 - 876176$
 $= 8521$.

$$\begin{array}{r} 884697 \\ - 876176 \\ \hline 8521 \end{array}$$

86. Let $10531 + 4813 - 728 = x \times 87$. Then
 $(15344 - 728) = 87 \times x \Rightarrow x = \frac{14616}{87} = 168$.

$$\begin{array}{r} 10531 \\ + 4813 \\ \hline 15344 \\ - 728 \\ \hline 14616 \end{array}$$

$$\begin{array}{r} 168 \\ 87 \overline{)14616} \\ \underline{87} \\ 591 \\ \underline{522} \\ 696 \\ \underline{696} \\ \hline \end{array}$$

87. $394 \times 113 = 394 \times (100 + 10 + 3)$
 $= (394 \times 100) + (394 \times 10) + (394 \times 3)$
 $= 39400 + 3940 + 1182$
 $= 44522$.

$$\begin{array}{r} 39400 \\ 3940 \\ + 1182 \\ \hline 44522 \end{array}$$

88. $1260 \div 14 \div 9 = \left(1260 \times \frac{1}{14} \times \frac{1}{9}\right) = 10$.

89. $136 \times 12 \times 8 = 136 \times 96 = 136 \times (100 - 4)$
 $= (136 \times 100) - (136 \times 4)$
 $= 13600 - 544 = 13056$.

$$\begin{array}{r} 13600 \\ - 544 \\ \hline 13056 \end{array}$$

90. Let $8888 + 848 + 88 - x = 7337 + 737$.
Then $9824 - x = 8074$